# Neelasha Sudarshan

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## EDUCATION

**Santa Clara University**, *Santa Clara, CA* *Expected June 2029*
**Bachelor of Science in Computer Science and Engineering** (GPA: 3.84)

- Relevant Coursework: Abstract Data Types and Data Structures; Embedded Systems; Linear Algebra

## WORK EXPERIENCE

**SPAR AI Research Fellow** [Code] *Sept 2026 - Current*

- Accepted to SPAR AI Research Fellowship (**14.9% acceptance**, 6k+ applications)
- Built config-based reinforcement learning pipeline (PPO on Tinker, DeepSeek V3.1) to elicit and red-team collusion pathways between AI models.

**AI Research Fellow** [Write-up, Code] *June 2026 - Current*
Santa Clara University, Department of Computer Science and Engineering, *Santa Clara, CA*

- Funded by SCU Undergraduate Research Award.
- Replicated passive-collusion results via LoRA fine-tuning of gpt-oss-20b (self-recognition ROC-AUC 0.98).
- Designed a causal belief-flip experiment testing whether self-recognition drives collusive monitor scores.

## PROJECTS

**White-Box Steering Safeguards for AI [Write-up, Code]**

- Tested activation steering strategies to lower prompt injection attack rates on gpt-oss-20b; reduced attack success from **93% to 3%**.
- Showed probe-gated steering fails despite an accurate probe (AUC 0.88). Key result: attack suppression requires steering full injection span.

**Open-Source Data Privacy Evaluation Framework [Code]**

- Contributed to SynEval by implementing stratified 3-fold cross-validation.
- Added PR-AUC scores so membership inference results are reliable for imbalanced datasets.

**MindGarden - Study Timer | Winner - "Best in Play: Top Game" - Hack for Humanity 2026 [DevPost]**

- Pitched virtual garden study timer concept to my 5-person team.
- Recruited a graphic designer and defined the aesthetic direction for the website.

## CERTIFICATIONS

**BlueDot Impact Technical AI Safety** *August 2026*

- Completed 30-hour program covering AI safety techniques, key gaps, and various threat models. Created a one-page action plan for contributing to AI Safety.

## LEADERSHIP

**Webmaster, SCU Association for Computing Machinery** *April 2026 - Current*

- Redesigned the club website's visual identity (color palette, logos, fonts) and maintained the site, fixing display bugs and updating board profiles.
- Delivered a technical demo at a Git/GitHub-based collaborative coding workshop for ~20 students.

## SKILLS

**Languages:** C, Python, HTML/CSS/JS
**AI/ML:** Hugging Face Transformers, PEFT/LoRA, Unsloth, Tinker, scikit-learn, Weights & Biases, SFT, RL (PPO), activation steering, linear probes, causal interventions, Membership Inference Attack (MIA)
**Web:** React, Tailwind, Vite, Firebase, UI/UX
**Applications:** GitHub, Docker, Inspect AI, RunPod
**Operating Systems:** macOS, Linux